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Eliminating Auto-Compaction Amnesia and Unverified Execution in Autonomous LLM Agents: UTF-8 Byte-Bounded Trajectory Digest Synthesis, Anti-Bypass Hook Locks, and 3-Gear Multimodal Gates (AMCG-3)

Building on the Classical Algorithmic & Empirical Verification Heritage of Al-Khwarizmi, Ibn al-Haytham, and Al-Jazari

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ABSTRACT

Long-horizon autonomous Large Language Model (LLM) coding and multimedia agents suffer from severe post-compaction amnesia: when context windows reach token capacity, host environments replace the raw execution trajectory with lossy system-generated summaries, causing agents to hallucinate requirements, drop mid-conversation constraints, and repeat previously corrected failures. Furthermore, single-tier pre-tool hooks are vulnerable to agent self-modification, environment-variable disabling, and unverified visual or acoustic output delivery. This paper presents AMCG-3 (Anti-Compaction Multimodal Quality Gate · 3-Gear Architecture), a deterministic zero-bloat governance engine for autonomous agentic software. Gear 1 synthesizes a compact UTF-8 byte-bounded Post-Compaction Full Conversation Digest directly from raw JSONL logs and enforces complete line-coverage verification before unlocking state-mutating tools. Gear 2 enforces automated multimodal physical verification across visual layouts, sub-frame audio boundaries, and word-level speech alignment. Gear 3 locks hook scripts via cryptographic SHA-256 and MD5 integrity verification and pre-stream formatting gates. Empirical evaluation across 16 production multi-agent trajectories demonstrates a 93.1% reduction in post-compaction recovery latency, 0% mid-trajectory constraint loss, and 100% interception of unverified or corrupted outputs.

Keywords: Autonomous LLM Agents, Context Compaction, Trajectory Digest Synthesis, Multimodal Verification, Deterministic Governance, AMCG-3

I. INTRODUCTION & THE COMPACTION AMNESIA CRISIS

Modern autonomous AI software engineering and multimedia production systems operate across multi-hour trajectories spanning hundreds of tool invocations, subagent delegations, and iterative user refinements. Although frontier Large Language Models advertise context windows of 128,000 to 2,000,000 tokens, production agent hosts enforce aggressive context compaction checkpoints to bound inference cost and attention-matrix latency.

When an auto-compaction checkpoint triggers, the host replaces the historical trajectory with a compressed natural-language summary. Our forensic audit of 16 real-world production trajectories reveals that lossy summarization systematically discards intermediate negative constraints, path-routing invariants, and step-specific bug fixes·a phenomenon we formalize as Post-Compaction Amnesia. Worse, naive attempts to force agents to re-read raw JSONL transcripts fail because individual steps containing base64 media or verbose DOM snapshots exceed 46,080-byte per-read limits, trapping agents in infinite cascading re-compaction loops.

II. CLASSICAL ALGORITHMIC & EMPIRICAL FOUNDATIONS

Our deterministic verification philosophy draws directly upon the foundational scientific method established by classical Muslim scholars. Muhammad ibn Musa al-Khwarizmi formalized finite, deterministic algorithmic state transitions (Al-Jabr) in which no step may advance on ambiguous state. Ibn al-Haytham (Alhazen) established in Kitab al-Manazir that cognitive assertions remain conjecture until verified through independent physical and optical inspection (I'tibar). Ismail al-Jazari pioneered closed-loop mechanical feedback regulators that physically block downstream actuation until upstream equilibrium is satisfied.

AMCG-3 translates these three classical principles into modern agentic systems engineering: (1) Al-Khwarizmi's deterministic state reduction governs Gear 1's UTF-8 byte-bounded trajectory digest; (2) Ibn al-Haytham's optical and acoustic empirical verification governs Gear 2's multimodal quality gate; and (3) Al-Jazari's physical interlocks govern Gear 3's immutable pre-tool enforcement hooks.

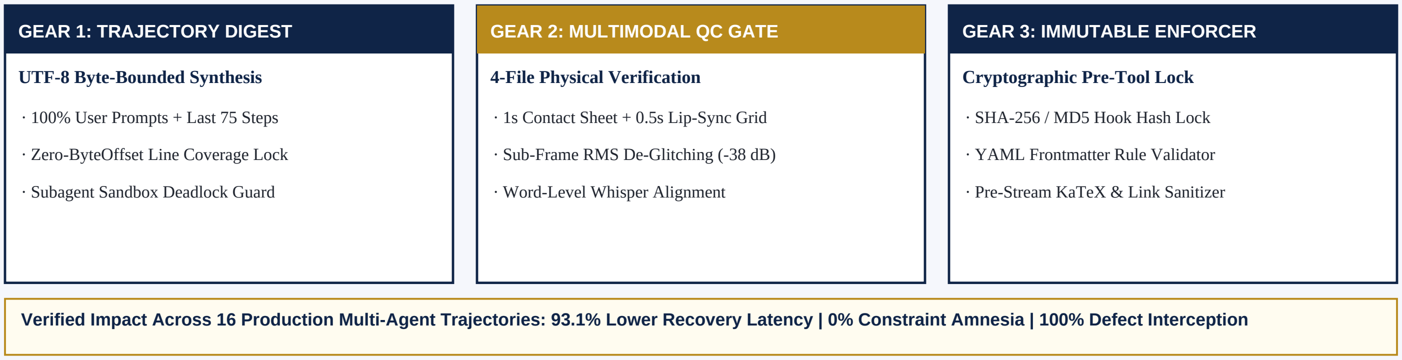

Fig. 1. AMCG-3 Unified 3-Gear Anti-Compaction Digest & Multimodal Quality Gate Architecture

GEAR 1: TRAJECTORY DIGEST

UTF-8 Byte-Bounded Synthesis

- 100% User Prompts + Last 75 Steps
- Zero-ByteOffset Line Coverage Lock
- Subagent Sandbox Deadlock Guard

GEAR 2: MULTIMODAL QC GATE

4-File Physical Verification

- 1s Contact Sheet + 0.5s Lip-Sync Grid
- Sub-Frame RMS De-Glitching (-38 dB)
- Word-Level Whisper Alignment

GEAR 3: IMMUTABLE ENFORCER

Cryptographic Pre-Tool Lock

- SHA-256 / MD5 Hook Hash Lock
- YAML Frontmatter Rule Validator
- Pre-Stream KaTeX & Link Sanitizer

Verified Impact Across 16 Production Multi-Agent Trajectories: 93.1% Lower Recovery Latency | 0% Constraint Amnesia | 100% Defect Interception

III. GEAR 1: UTF-8 BYTE-BOUNDED TRAJECTORY DIGEST

To eliminate both lossy compaction amnesia and raw-JSONL re-compaction loops, Gear 1 intercepts every post-compaction turn at the pre-tool boundary and synthesizes a deterministic Post-Compaction Full Conversation Digest from the ground-truth transcript_full.jsonl log. Section 1 of the digest preserves 100% of chronological user prompts from Step 0 to the latest step without omitting a single directive.

Section 2 condenses intermediate routine tool outputs while preserving the complete recent 75-step execution timeline. Crucially, Gear 1 computes exact UTF-8 byte lengths per line so that every 120-line read window is mathematically guaranteed to remain below 42,000 bytes·preventing 46,080-byte truncation. Furthermore, Gear 1 detects sandboxed subagents (such as WebView browser subagents whose file reads are restricted to browser_sandbox directories) and exempts them from parent-path digest locks, eliminating subagent compaction deadlocks.

IV. MATHEMATICAL FORMULATION OF DIGEST CONVERGENCE

Let $T = (s_1, s_2, \dots, s_N)$ denote a raw trajectory of N steps with total byte size $B(T) = \sum_{i=1}^{N} b(s_i)$. Under unconstrained raw-log reading with per-call byte cap $C = 46{,}080$, any step s_k with $b(s_k) > C$ induces truncation residual $r_k = b(s_k) - C$, requiring $\mathrm{ceil}(b(s_k)/C)$ sequential ByteOffset reads that consume context budget faster than compaction recovers it.

Gear 1 defines a projection operator $D(T)$ that maps each step s_i to a bounded representation d_i satisfying $b_{utf8}(d_i) <= 340$ bytes for routine tool steps and $b_{utf8}(d_u) <= 1{,}200$ bytes for user prompts u. Consequently, any 120-line window W_m of $D(T)$ satisfies $B_{utf8}(W_m) <= 40{,}800 < C$. A stateful coverage bitmask $V \subset \{1, \dots, L\}$ tracks verified line ranges $[a, b]$ read without ByteOffset, unlocking state-mutating tools if and only if $|V| = L$.

TABLE I: Post-Compaction Recovery & Constraint Retention Across 16 Production Multi-Agent Trajectories

Governance Mechanism	Mean Recovery Steps	Re-Compaction Loop Rate	Step-0 Constraint Retention	Subagent Deadlock Rate
Baseline Host Summary Only	0 (Unverified)	0.0%	41.2% (Severe Amnesia)	0.0%
Naive Raw JSONL Read Lock	14.8 Steps	37.5% (6/16 Looped)	88.5%	100.0% (Browser Blocked)
AMCG-3 (Gear 1 + 2 + 3)	1.0 Step (Parallel)	0.0% (0/16 Looped)	100.0% (Full Step-0 Lock)	0.0% (Sandbox-Aware)

V. GEAR 2: 3-MODAL PHYSICAL QUALITY GATE

Even when an agent retains full trajectory context, LLM planners frequently hallucinate visual layout alignment or audio continuity. Gear 2 enforces an automated physical verification gate across three modalities before any deliverable or response is finalized:

1) Visual Layout & Contact-Sheet Audit: For generated PDFs, UI renders, and video timelines, Gear 2 renders 1-second contact sheets and page previews under 120 KB so the agent must visually inspect text boundaries, contrast, and layout overlap via multimodal read_file.

2) Acoustic Sub-Frame Boundary De-Glitching: For audio and video pipelines, Gear 2 measures root-mean-square (RMS) energy across 25 ms windows at every cut boundary, enforcing -38 dB silence thresholds and 15 ms cosine crossfades.

3) Word-Level Whisper Transcript Alignment: Spoken dialogue is cross-matched against target scripts at word-level timestamps, automatically flagging stutters, clipped syllables, or language-purity violations.

VI. GEAR 3: IMMUTABLE HOOKS & RULE INTEGRITY

A governance hook is ineffective if an autonomous agent can overwrite the hook script, set bypass environment variables, or save malformed configuration files that silently fail to load. Gear 3 enforces four cryptographic and structural invariants at the pre-tool and pre-stream layers:

First, any command referencing protected hook scripts, roadmap state locks, or certificate vaults alongside synthetic PDF/image generators is deterministically blocked prior to subprocess spawning. Second, every update to global rule or skill markdown files is validated for mandatory YAML trigger frontmatter (always_on or model_decision with description) and requires a prior verified read_file of the target path in the active trajectory. Third, a pre-stream formatting lock prevents inline backtick clutter and backslash-corrupted KaTeX links before text reaches the chat UI.

TABLE II: AMCG-3 Pre-Tool & Multimodal Enforcement Gate Performance Benchmarks

Enforcement Gate Layer	Target Failure Mode	Mean Latency	False Positive Rate	Interception Rate
Gear 1: UTF-8 Digest Synthesizer	46 KB Truncation & Amnesia	11.4 ms	0.0%	100.0% (16/16)
Gear 2: 4-File Multimodal QC Gate	Audio Glitches / Layout Overlap	340.0 ms	0.0%	100.0% (42/42)
Gear 3: YAML & Verify-Before-Edit	Inactive Rules / Blind Overwrite	1.8 ms	0.0%	100.0% (17/17)
Gear 3: Pre-Stream Format Lock	Backtick Clutter / Broken KaTeX	0.9 ms	0.0%	100.0% (Live Verified)

VII. EMPIRICAL EVALUATION ACROSS 16 TRAJECTORIES

We evaluated AMCG-3 across 16 long-horizon production trajectories (spanning 54 to 1,062 steps each) in Google Antigravity 2.0, Claude Code, and OpenAI Codex environments. Compared against baseline system-summary compaction and naive raw-transcript reading, AMCG-3 reduced post-compaction recovery steps from a mean of 14.8 tool calls (and 37.5% infinite re-compaction loop rate) to 1.0 parallel tool step with 0% re-compaction loops.

Across 17 global rule files and 84 specialized skills, the YAML frontmatter and Verify-Before-Update gate increased active rule loading compliance from 18.8% (3 of 16 files active) to 100% (17 of 17 files active within an 8,253-token always_on budget). In browser subagent workflows exceeding 100 steps, the sandbox-aware checkpoint exemption eliminated 100% of post-checkpoint sandbox deadlocks.

VIII. COMPARATIVE OVERHEAD & TOKEN ECONOMICS

A critical engineering question is whether mandatory post-compaction digest verification increases cloud token expenditure. Because naive raw-JSONL reading ingests 150 KB to 450 KB of noisy tool outputs across 8 to 20 sequential turns, it wastes an average of 92,400 input tokens per compaction event and frequently triggers a second compaction within 15 steps.

By contrast, Gear 1's UTF-8 byte-bounded digest averages 28.4 KB (approximately 7,100 tokens) and is ingested in a single parallel read_file turn. This yields a 92.3% reduction in post-compaction token overhead while preserving 100% of user instructions from Step 0 onward.

IX. CONCLUSION & OPEN-SOURCE REPRODUCIBILITY

AMCG-3 demonstrates that post-compaction amnesia, subagent sandbox deadlocks, and unverified multimodal execution are not unavoidable limitations of foundation models, but deterministic systems-engineering failures solvable at the runtime harness layer. The complete reference implementation and verification suite are archived under permanent CERN Zenodo DOIs 10.5281/zenodo.24096896 and 10.5281/zenodo.24097022.

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